

Narrative approach and local projection

Qingyuan Fang¹

¹Johns Hopkins University

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Overview

Big picture

We want an empirical object that behaves like a **policy innovation**:

$$\text{monetary policy shock}_t \approx \Delta i_t \perp\!\!\!\perp \mathcal{I}_t,$$

where $\mathcal{I}_t$ is the information set that drives systematic policy.

- Identification problem: **endogeneity** and **anticipation**.
- The narrative approach in Romer and Romer (2004):
 - Step 1: intended policy change around FOMC meetings.
 - Step 2: purge systematic responses to Fed forecasts.
- Implications: output & prices; why results differ from conventional measures.
- Bridge to **local projections** and what a modern LP replication would change.

Two core threats

Endogeneity (policy responds to the economy)

In booms, rates rise endogenously $\Rightarrow$ attenuate the contractionary estimate.

Anticipation (policy responds to Fed forecasts)

Even a “target” series embeds forward-looking responses, eg.

$$i_t = \phi \mathbb{E}_t[\pi_{t+1:t+h}, y_{t+1:t+h}] + \varepsilon_t.$$

If the Fed eases because it foresees a recession, output may fall *after* easing even when easing is stimulative.

Identification target: “deviations from the systematic rule”

What we want

A shock should be orthogonal to the information set driving the systematic component:

$$\varepsilon_t \perp\!\!\!\perp \mathcal{I}_t \quad \Rightarrow \quad \mathbb{E}[\varepsilon_t | \mathcal{I}_t] = 0.$$

What makes this hard in monetary policy

- $\mathcal{I}_t$ is huge: forecasts, judgment, informal signals, financial conditions, global news.
- Identification strategies differ mainly in **how they approximate $\mathcal{I}_t$** and **when** they measure the surprise.

Preview: two complementary “information sets”

- Narrative approach: approximates *policy maker* information set.
- High-frequency approach: approximates *market* information set in a narrow window.

Why focus on FOMC meetings?

Key constraint

The staff forecast used as the proxy for $\mathcal{J}_t$ is produced **for scheduled FOMC meetings**.

T-6 days

Greenbook forecast; used heavily in deliberations.

T-5 to T

Bluebook alternatives; pre-meeting briefings; incoming data possibly triggers forecast update.

Meeting day

FOMC decisions shape intended stance.

Post-meeting

Desk implements operations; intermeeting moves may occur on new information.

Design choice

Concentrate on **policy intentions** over the window where the forecast is a good proxy.

Data Sources

Narrative sources

- Record of Policy Actions.
- Detailed meeting accounts.
- Bluebooks (Monetary Policy Alternatives): implementation + options.

Why narrative?

- Extends policy-intention measurement to periods with less explicit targets.
- Lets we distinguish: *policy decided at meeting vs reaction to new information*.

Quantitative internal source

Internal memos reporting the “**expected federal funds rate**” (Weekly Report of the Manager of Open Market Operations), used as an input/check for intended levels and changes.

Why focus on the funds rate?

- Explicit target for large parts of sample.
- Discussed even when reserves were emphasized.
- More consistent over time than quantity aggregates.

Step 1: Intended funds rate change around each meeting

Meeting-level object

For meeting m , infer two intended levels:

$ff_m^{\text{pre}} \equiv$ intended funds rate at forecast date, $ff_m^{\text{post}} \equiv$ intended funds rate implied by meeting decision.

Then

$$\Delta ff_m^I \equiv ff_m^{\text{post}} - ff_m^{\text{pre}}.$$

- Use narrative record to adjust for temporary deviations from intended levels.
- Combine narrative + expected-rate memos when timing is ambiguous:
 - Some near-post-meeting moves are *new information*; others were *pre-decided*.

Step 2

High-level specification (meeting-level)

$$\Delta ff_m^I = \alpha + \rho ff_m^{\text{pre}} + \underbrace{g(\tilde{y}_{m,h}, \Delta \tilde{y}_{m,h}, \tilde{\pi}_{m,h}, \Delta \tilde{\pi}_{m,h}, \tilde{u}_{m,0})}_{\text{Fed forecast controls}} + \varepsilon_m.$$

What goes into $g(\cdot)$ in the baseline

- Real output growth forecasts at horizons $h = -1, 0, 1, 2$ and changes since previous meeting.
- Inflation (deflator) forecasts at same horizons and changes since previous meeting.
- Current-quarter unemployment forecast.

Definition of the shock and how to interpret the residual

Shock definition

$$S_m \equiv \varepsilon_m \quad \Rightarrow \quad S_t \equiv \sum_{m \in \text{meetings in month } t} \varepsilon_m.$$

Interpret S_t as the change in intended policy not explained by forecast-based systematic responses.

Residuals are not “random mistakes”

Any regression-based shock absorbs **everything** not captured by the specification:

- operating procedures; beliefs; preferences; political pressure; other objectives; idiosyncratic factors;
- **plus** any missing information (if Greenbook is incomplete).

Sources of variation in S_t

- **Operating procedures:** reserve targeting episodes shift rates relative to forecast-implied rule.
- **Beliefs:** model of inflation-output tradeoff, NAIRU views, etc.
- **Preferences and tastes:** stronger inflation aversion $\Rightarrow$ higher rate for given forecast.
- **Politics:** pressure episodes can move the stance.
- **Other objectives:** e.g., exchange-rate concerns beyond forecast implications.
- **Idiosyncratic randomness:** personalities, persuasion, etc.

Effect on industrial production (Eq. 2)

$$\Delta y_t = a_0 + \sum_{k=1}^{11} a_k D_{k,t} + \sum_{i=1}^{24} b_i \Delta y_{t-i} + \sum_{j=1}^{36} c_j S_{t-j} + e_t.$$

Key design choices

- Use **non-seasonally adjusted** IP $\Rightarrow$ include month dummies $D_{k,t}$.
- Include many lags of Δy_t (normal dynamics) and S_t (policy lag structure).
- Exclude contemporaneous S_t : assume monetary policy affects output with at least a one-month lag.

How the impulse response is computed

A one-time 1pp realization of S maps into a path for Δy via the c_j and feedback through b_i ; cumulate to get level effect on y .

Effect on prices (Eq. 3)

$$\Delta p_t = \alpha_0 + \sum_{k=1}^{11} \alpha_k D_{k,t} + \sum_{i=1}^{24} \beta_i \Delta p_{t-i} + \sum_{j=1}^{48} \gamma_j S_{t-j} + u_t.$$

Key takeaways

- Estimated price-level response is **near zero for a long time** (order of 2 years), then disinflation.
- Using broad measures like Δ funds rate can yield a **price puzzle** (prices rise after tightening).
- Adding commodity prices helps in some settings, but is not a complete fix if policy remains endogenous/anticipatory.

Why both corrections matter: “intended” and “forecast-purged”

Four instruments we should conceptually separate

- ① Δi_t (*actual* funds rate change): mixes policy, noise, intermeeting info responses.
- ② Δi_t^I (*intended* change around meetings): reduces short-run endogeneity, but still anticipatory.
- ③ residual from regressing Δi_t on forecasts: partial anticipation control, but keeps short-run endogeneity.
- ④ **New measure**: intended changes *and* forecast purge.

Empirical punchline to emphasize

Intermediate measures tend to look more like the broad measure than like the new measure
⇒ we need to address **both** endogeneity and anticipation to change estimated transmission meaningfully.

From Eq. (2) to local projections

Eq. (2) estimates one equation for Δy_t with long lag polynomials, then **derives** a multi-horizon response by iterating dynamics

Local projections (LP): horizon-by-horizon direct estimation

For each horizon $h = 0, 1, \dots, H$ estimate a separate regression:

$$\Delta y_{t+h} = \alpha_h + \theta_h S_t + \Gamma'_h W_t + v_{t+h},$$

where W_t collects predetermined controls (lags of y , lags of S , and other covariates).

Narrative approach

More narrative datasets, clearer “instrument” framing

Narrative shocks are often best viewed as **instruments** (deviations from policy rules) used to trace causal propagation.

Scale narrative coding with text methods

- Use natural language from internal policy documents to proxy a broader information set than a few forecast series.
- Apply structured text/ML pipelines to extract and orthogonalize rich information embedded in documents.

Further Reading

Jordà, Ò. (2005). “Estimation and Inference of Impulse Responses by Local Projections.” *American Economic Review*, 95(1), 161–182.

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Drechsel, T. (forthcoming). “Political Pressure on the Fed.” *Review of Economic Studies*.

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Martingale Difference Sequence (MDS)

A MDS e_t is defined with respect to a specific sequence of **information sets** $\mathcal{F}_t$.

Definition

The process $(e_t, \mathcal{F}_t)$ is a MDS if

- e_t is adapted to $\mathcal{F}_t$, meaning e_t is $\mathcal{F}_t$ -measurable for all t .
- $\mathbb{E}|e_t| < \infty$
- $\mathbb{E}[e_t | \mathcal{F}_{t-1}] = 0$

Note: When no explicit definition is given it's standard to assume that $\mathcal{F}_t = \sigma(e_t, e_{t-1}, \dots)$.

- ① If e_t is i.i.d and mean zero, it's a MDS
- ② If $(e_t, \mathcal{F}_t)$ is a MDS and $\mathbb{E}[e_t^2] < \infty$ then e_t is serially uncorrelated.

Ex. Show that if $(e_t, \mathcal{F}_t)$ is a MDS and X_t is $\mathcal{F}_t$ -measurable then $u_t = X_{t-1}e_t$ is a MDS